# 8 QUADRATIC EQUATIONS

## Exercise-1 : Single Choice Problems

1. Sum of values of $x$ and $y$ satisfying the equation $3^x - 4^y = 77;\ 3^{x/2} - 2^y = 7$ is :
   (a) 2 (b) 3 (c) 4 (d) 5

2. If $f(x) = \prod_{i=1}^{3}(x - a_i) + \sum_{i=1}^{3} a_i - 3x$ where $a_i < a_{i+1}$ for $i = 1, 2$, then $f(x) = 0$ has :
   (a) only one distinct real root (b) exactly two distinct real roots
   (c) exactly 3 distinct real roots (d) 3 equal real roots

3. Complete set of real values of '$a$' for which the equation $x^4 - 2ax^2 + x + a^2 - a = 0$ has all its roots real :
   (a) $\left[\frac{3}{4}, \infty\right)$ (b) $[1, \infty)$ (c) $[2, \infty)$ (d) $[0, \infty)$

4. The cubic polynomial with leading coefficient unity all whose roots are 3 units less than the roots of the equation $x^3 - 3x^2 - 4x + 12 = 0$ is denoted as $f(x)$, then $f'(x)$ is equal to:
   (a) $3x^2 - 12x + 5$ (b) $3x^2 + 12x + 5$ (c) $3x^2 + 12x - 5$ (d) $3x^2 - 12x - 5$

5. The set of values of $k\,(k \in R)$ for which the equation $x^2 - 4|x| + 3 - |k - 1| = 0$ will have exactly four real roots, is :
   (a) $(-2, 4)$ (b) $(-4, 4)$ (c) $(-4, 2)$ (d) $(-1, 0)$

6. The number of integers satisfying the inequality $\frac{x}{x+6} \le \frac{1}{x}$ is :
   (a) 7 (b) 8 (c) 9 (d) 3

7. The product of uncommon real roots of the two polynomials $p(x) = x^4 + 2x^3 - 8x^2 - 6x + 15$ and $q(x) = x^3 + 4x^2 - x - 10$ is:
   (a) 4 (b) 6 (c) 8 (d) 12

8. If $\lambda_1, \lambda_2\ (\lambda_1 > \lambda_2)$ are two values of $\lambda$ for which the expression $f(x, y) = x^2 + \lambda xy + y^2 - 5x - 7y + 6$ can be resolved as a product of two linear factors, then the value of $3\lambda_1 + 2\lambda_2$ is :
   (a) 5 (b) 10 (c) 15 (d) 20

**9.** Let $\alpha, \beta$ be the roots of the quadratic equation $ax^2 + bx + c = 0$, then the roots of the equation $a(x+1)^2 + b(x+1)(x-2) + c(x-2)^2 = 0$ are :

(a) $\frac{2\alpha+1}{\alpha-1}, \frac{2\beta+1}{\beta-1}$
(b) $\frac{2\alpha-1}{\alpha+1}, \frac{2\beta-1}{\beta+1}$
(c) $\frac{\alpha+1}{\alpha-2}, \frac{\beta+1}{\beta-2}$
(d) $\frac{2\alpha+3}{\alpha-1}, \frac{2\beta+3}{\beta-1}$

**10.** If $a, b \in R$ distinct numbers satisfying $|a-1|+|b-1|=|a|+|b|=|a+1|+|b+1|$, then the minimum value of $|a-b|$ is :

(a) 3 (b) 0 (c) 1 (d) 2

**11.** The smallest positive integer $p$ for which expression $x^2 - 2px + 3p + 4$ is negative for atleast one real $x$ is :

(a) 3 (b) 4 (c) 5 (d) 6

**12.** For $x \in R$, the expression $\frac{x^2+2x+c}{x^2+4x+3c}$ can take all real values if $c \in$:

(a) $(1, 2)$
(b) $[0, 1]$
(c) $(0, 1)$
(d) $(-1, 0)$

**13.** If 2 lies between the roots of the equation $t^2 - mt + 2 = 0$ , $(m \in R)$ then the value of $\left[\left(\frac{3|x|}{9+x^2}\right)^m\right]$ is :

(where $[\cdot]$ denotes greatest integer function)

(a) 0 (b) 1 (c) 8 (d) 27

**14.** The number of integral roots of the equation $x^8 - 24x^7 - 18x^5 + 39x^2 + 1155 = 0$ is :

(a) 0 (b) 2 (c) 4 (d) 6

**15.** If the value of $m^4 + \frac{1}{m^4} = 119$, then the value of $\left|m^3 - \frac{1}{m^3}\right| =$

(a) 11 (b) 18 (c) 24 (d) 36

**16.** If the equation $ax^2 + 2bx + c = 0$ and $ax^2 + 2cx + b = 0$, $a \neq 0$, $b \neq c$, have a common root, then their other roots are the roots of the quadratic equation:

(a) $a^2x(x+1) + 4bc = 0$
(b) $a^2x(x+1) + 8bc = 0$
(c) $a^2x(x+2) + 8bc = 0$
(d) $a^2x(1+2x) + 8bc = 0$

**17.** If $\cos\alpha$, $\cos\beta$ and $\cos\gamma$ are the roots of the equation $9x^3 - 9x^2 - x + 1 = 0$; $\alpha, \beta, \gamma \in [0, \pi]$ then the radius of the circle whose centre is $(\Sigma\alpha, \Sigma\cos\alpha)$ and passing through $(2\sin^{-1}(\tan\pi/4), 4)$ is:

(a) 2 (b) 3 (c) 4 (d) 5

**18.** For real values of $x$, the value of expression $\frac{11x^2-12x-6}{x^2+4x+2}$:

(a) lies between $-17$ and $-3$
(b) does not lie between $-17$ and $-3$
(c) lies between 3 and 17
(d) does not lie between 3 and 17

**19.** $\dfrac{x+3}{x^2-x-2} \geq \dfrac{1}{x-4}$ holds for all $x$ satisfying:
(a) $-2 < x < 1$ or $x > 4$
(b) $-1 < x < 2$ or $x > 4$
(c) $x < -1$ or $2 < x < 4$
(d) $x > -1$ or $2 < x < 4$

**20.** If $x = 4 + 3i$ (where $i = \sqrt{-1}$), then the value of $x^3 - 4x^2 - 7x + 12$ equals:
(a) $-88$ (b) $48 + 36i$ (c) $-256 + 12i$ (d) $-84$

**21.** Let $f(x) = \dfrac{x^2+x-1}{x^2-x+1}$, then the largest value of $f(x)\ \forall x \in [-1, 3]$ is:
(a) $\dfrac{3}{5}$ (b) $\dfrac{5}{3}$ (c) 1 (d) $\dfrac{4}{3}$

**22.** In above problem, the range of $f(x) \forall x \in [-1, 1]$ is:
(a) $\left[-1, \dfrac{3}{5}\right]$ (b) $\left[-1, \dfrac{5}{3}\right]$ (c) $\left[-\dfrac{1}{3}, 1\right]$ (d) $[-1, 1]$

**23.** If the roots of the equation $\dfrac{1}{x+p} + \dfrac{1}{x+q} = \dfrac{1}{r}$ are equal in magnitude but opposite in sign, then the product of the roots is:
(a) $-2(p^2+q^2)$ (b) $-(p^2+q^2)$ (c) $-\dfrac{(p^2+q^2)}{2}$ (d) $-pq$

**24.** If a root of the equation $a_1x^2 + b_1x + c_1 = 0$ is the reciprocal of a root of the equation $a_2x^2 + b_2x + c_2 = 0$, then :
(a) $(a_1a_2 - c_1c_2)^2 = (a_1b_2 - b_1c_2)(a_2b_1 - b_2c_1)$
(b) $(a_1a_2 - b_1b_2)^2 = (a_1b_2 - b_1c_2)(a_2b_1 - b_2c_1)$
(c) $(b_1c_2 - b_2c_1)^2 = (a_1b_2 - b_1c_2)(a_2b_1 + b_2c_1)$
(d) $(b_1c_2 - b_2c_1)^2 = (a_1b_2 + b_1c_2)(a_2b_1 - b_2c_1)$

**25.** If $\alpha \neq \beta$ but $\alpha^2 = 5\alpha - 3$ and $\beta^2 = 5\beta - 3$, then the equation with roots $\dfrac{\alpha}{\beta}, \dfrac{\beta}{\alpha}$ is:
(a) $3x^2 - 25x + 3 = 0$
(b) $x^2 + 5x - 3 = 0$
(c) $x^2 - 5x + 3 = 0$
(d) $3x^2 - 19x + 3 = 0$

**26.** If the difference between the roots of $x^2 + ax + b = 0$ is same as that of $x^2 + bx + a = 0$, $a \neq b$, then:
(a) $a + b + 4 = 0$ (b) $a + b - 4 = 0$ (c) $a - b - 4 = 0$ (d) $a - b + 4 = 0$

**27.** If $\tan\theta_i$; $i = 1, 2, 3, 4$ are the roots of equation $x^4 - x^3 \sin 2\beta + x^2 \cos 2\beta - x\cos\beta - \sin\beta = 0$, then $\tan(\theta_1 + \theta_2 + \theta_3 + \theta_4) =$
(a) $\sin\beta$ (b) $\cos\beta$ (c) $\tan\beta$ (d) $\cot\beta$

**28.** Let $a$, $b$, $c$, $d$ are positive real numbers such that $\frac{a}{b} \neq \frac{c}{d}$, then the roots of the equation: $(a^2+b^2)x^2+2x(ac+bd)+(c^2+d^2)=0$ are:

(a) real and distinct
(b) real and equal
(c) imaginary
(d) nothing can be said

**29.** If $\alpha$, $\beta$ are the roots of $ax^2+bx+c=0$, then the equation whose roots are $2+\alpha$, $2+\beta$, is :

(a) $ax^2+x(4a-b)+4a-2b+c=0$
(b) $ax^2+x(4a-b)+4a+2b+c=0$
(c) $ax^2+x(b-4a)+4a+2b+c=0$
(d) $ax^2+x(b-4a)+4a-2b+c=0$

**30.** Minimum possible number of positive root of the quadratic equation $x^2-(1+\lambda)x+\lambda-2=0$, $\lambda \in R$ :

(a) 2
(b) 0
(c) 1
(d) can not be determined

**31.** Let $\alpha$, $\beta$ be real roots of the quadratic equation $x^2+kx+(k^2+2k-4)=0$, then the minimum value of $\alpha^2+\beta^2$ is equal to :

(a) 12 (b) $\frac{4}{9}$ (c) $\frac{16}{9}$ (d) $\frac{8}{9}$

**32.** Polynomial $P(x)=x^2-ax+5$ and $Q(x)=2x^3+5x-(a-3)$ when divided by $x-2$ have same remainders, then '$a$' is equal to:

(a) 10 (b) $-10$ (c) 20 (d) $-20$

**33.** If $a$ and $b$ are non-zero distinct roots of $x^2+ax+b=0$, then the least value of $x^2+ax+b$ is equal to :

(a) $\frac{2}{3}$ (b) $\frac{9}{4}$ (c) $-\frac{9}{4}$ (d) 1

**34.** Let $\alpha, \beta$ be the roots of the equation $ax^2+bx+c=0$. A root of the equation $a^3x^2+abcx+c^3=0$ is :

(a) $\alpha+\beta$ (b) $\alpha^2+\beta$ (c) $\alpha^2-\beta$ (d) $\alpha^2\beta$

**35.** Let $a, b, c$ be the lengths of the sides of a triangle (no two of them are equal) and $k \in R$. If the roots of the equation $x^2+2(a+b+c)x+6k(ab+bc+ca)=0$ are real, then:

(a) $k<\frac{2}{3}$ (b) $k>\frac{2}{3}$ (c) $k>1$ (d) $k<\frac{1}{4}$

**36.** Root(s) of the equation $9x^2-18|x|+5=0$ belonging to the domain of definition of the function $f(x)=\log(x^2-x-2)$ is/are:

(a) $\frac{-5}{3}, \frac{-1}{3}$ (b) $\frac{5}{3}, \frac{1}{3}$ (c) $\frac{-5}{3}$ (d) $\frac{-1}{3}$

**37.** If $\beta+\cos^2\alpha, \beta+\sin^2\alpha$ are the roots of $x^2+2bx+c=0$ and $\gamma+\cos^4\alpha$, $\gamma+\sin^4\alpha$ are the roots of $x^2+2Bx+C=0$, then:

(a) $b-B=c-C$ (b) $b^2-B^2=c-C$ (c) $b^2-B^2=4(c-C)$ (d) $4(b^2-B^2)=c-C$

**38.** Minimum value of $|x-p|+|x-15|+|x-p-15|$. If $p \le x \le 15$ and $0 < p < 15$ :
(a) 30 (b) 15 (c) 10 (d) 0

**39.** If the quadratic equation $4x^2 - 2x - m = 0$ and $4p(q-r)x^2 - 2q(r-p)x + r(p-q) = 0$ have a common root such that second equation has equal roots then the value of $m$ will be :
(a) 0 (b) 1 (c) 2 (d) 3

**40.** The range of $k$ for which the inequality $k\cos^2 x - k\cos x + 1 \ge 0 \forall x \in (-\infty, \infty)$ is :
(a) $k > -\frac{1}{2}$ (b) $k > 4$ (c) $-\frac{1}{2} \le k \le 4$ (d) $\frac{1}{2} \le k \le 5$

**41.** If $\frac{1+\alpha}{1-\alpha}, \frac{1+\beta}{1-\beta}, \frac{1+\gamma}{1-\gamma}$ are roots of the cubic equation $f(x) = 0$ where $\alpha, \beta, \gamma$ are the roots of the cubic equation $3x^3 - 2x + 5 = 0$, then the number of negative real roots of the equation $f(x) = 0$ is:
(a) 0 (b) 1 (c) 2 (d) 3

**42.** The sum of all integral values of $\lambda$ for which $(\lambda^2 + \lambda - 2)x^2 + (\lambda + 2)x < 1 \, \forall x \in R$, is :
(a) $-1$ (b) $-3$ (c) 0 (d) $-2$

**43.** If $\alpha, \beta, \gamma, \delta \in R$ satisfy $\frac{(\alpha+1)^2 + (\beta+1)^2 + (\gamma+1)^2 + (\delta+1)^2}{\alpha+\beta+\gamma+\delta} = 4$

If biquadratic equation $a_0x^4 + a_1x^3 + a_2x^2 + a_3x + a_4 = 0$ has the roots $\left(\alpha + \frac{1}{\beta} - 1\right), \left(\beta + \frac{1}{\gamma} - 1\right), \left(\gamma + \frac{1}{\delta} - 1\right), \left(\delta + \frac{1}{\alpha} - 1\right)$. Then the value of $a_2/a_0$ is :
(a) 4 (b) $-4$ (c) 6 (d) none of these

**44.** If the complete set of value of $x$ satisfying $|x-1|+|x-2|+|x-3| \ge 6$ is $(-\infty, a] \cup [b, \infty)$, then $a + b =$:
(a) 2 (b) 3 (c) 6 (d) 4

**45.** If exactly one root of the quadratic equation $x^2 - (a+1)x + 2a = 0$ lies in the interval $(0, 3)$, then the set of value '$a$' is given by:
(a) $(-\infty, 0) \cup (6, \infty)$ (b) $(-\infty, 0] \cup (6, \infty)$
(c) $(-\infty, 0] \cup [6, \infty)$ (d) $(0, 6)$

**46.** The condition that the root of $x^3 + 3px^2 + 3qx + r = 0$ are in H.P. is :
(a) $2p^3 - 3pqr + r^2 = 0$ (b) $3p^3 - 2pqr + p^2 = 0$
(c) $2q^3 - 3pqr + r^2 = 0$ (d) $r^3 - 3pqr + 2q^3 = 0$

**47.** If $x$ is real and $4y^2 + 4xy + x + 6 = 0$, then the complete set of values of $x$ for which $y$ is real, is:
(a) $x \le -2$ or $x \ge 3$ (b) $x \le 2$ or $x \ge 3$ (c) $x \le -3$ or $x \ge 2$ (d) $-3 \le x \le 2$

**48.** The solution of the equation $\log_{\cos x^2}(3-2x) < \log_{\cos x^2}(2x-1)$ is :
(a) $(1/2, 1)$ (b) $(-\infty, 1)$
(c) $(1/2, 3)$ (d) $(1, \infty) - \sqrt{2n\pi}, \; n \in N$